

# Sentinel-2 L1C Time Series: Data Description

Khalil Bibi

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One acquisition of one site is one compressed NumPy file, `<YYYY-MM-DD>_<tile>.npz`. All acquisitions of a site share the same grid, so they stack directly into a time series. Every granule the catalogue returned — saved or rejected — is documented in `metadata.csv` next to the files.

## 1 Contents of one file

| Key                                             | Shape / type                         | Content                                                                                    |
|-------------------------------------------------|--------------------------------------|--------------------------------------------------------------------------------------------|
| <code>data</code>                               | (13, 1024, 1024) <code>uint16</code> | Top-of-atmosphere reflectance, raw ESA digital numbers                                     |
| <code>valid_mask</code>                         | (13, 1024, 1024) <code>bool</code>   | Per band: the pixel is usable                                                              |
| <code>nodata_mask</code>                        | (13, 1024, 1024) <code>bool</code>   | Per band: no sensor sample backs the pixel                                                 |
| <code>valid_any</code> / <code>valid_all</code> | (1024, 1024) <code>bool</code>       | Valid in at least one / in all 13 bands                                                    |
| <code>quality_bits</code>                       | (13, 1024, 1024) <code>uint8</code>  | ESA <code>MSK_QUALIT</code> flags, one bit per layer<br>( <code>quality_bit_names</code> ) |
| <code>classification_mask</code>                | (3, 1024, 1024) <code>uint8</code>   | ESA <code>MSK_CLASSI</code> : opaque cloud, cirrus, snow/ice                               |
| <code>bands</code>                              | (13,) <code>string</code>            | Band order of axis 0 (see below)                                                           |
| <code>transform, crs</code>                     | (6,) <code>float, string</code>      | Affine geotransform and CRS, e.g. <code>EPSG:32631</code>                                  |

Band order: B01, B02, B03, B04, B05, B06, B07, B08, B8A, B09, B10, B11, B12 — the 13 MSI bands from 443 nm (coastal aerosol) to 2190 nm (SWIR).

## 2 Geometry

The footprint is a  $1024 \times 1024$  pixel square at 10 m, i.e.  $10.24 \times 10.24$  km, in the UTM zone of the requested point (metric, so pixels are square everywhere). Its upper-left corner is snapped to a multiple of 60 m, which is the lattice shared by the 10, 20 and 60 m Sentinel-2 grids; the grid centre therefore sits a few tens of metres from the requested coordinate (23 m in the reference scene). The same grid is reused for every date of the site, so `[:, i, j]` is the same 10 m ground cell in every band and at every date.

| Native resolution | Bands                        | Processing                                                             |
|-------------------|------------------------------|------------------------------------------------------------------------|
| 10 m              | B02, B03, B04, B08           | Copied 1:1, no interpolation, when the granule is in the same UTM zone |
| 20 m              | B05, B06, B07, B8A, B11, B12 | Bilinear interpolation to 10 m                                         |
| 60 m              | B01, B09, B10                | Bilinear interpolation to 10 m                                         |

Resampling never invents data across gaps: a target pixel is valid only if every source sample in its interpolation kernel is valid, so pixels bordering a no-data area are flagged invalid rather than interpolated from partial support. Granules from a neighbouring UTM zone are reprojected, and then the 10 m bands are resampled too (the `source_crs` column says which case applies).

## 3 Radiometry

Values are Level-1C top-of-atmosphere reflectance, *not* atmospherically corrected (that would be Level-2A). Since processing baseline 04.00 — which covers the whole reprocessed archive — ESA adds an offset of 1000:

$$\rho_{\text{TOA}} = \frac{\text{DN} - 1000}{10000}.$$

Files store raw DN by default and record `processing_baseline`; `-harmonize` subtracts the offset at write time and sets the `harmonized` column. DN 0 means no data and DN 65535 marks saturation.

## 4 Validity and quality control

Each pixel is checked against, in order: the source no-data value, the interpolation-support rule above, and three ESA masks — `MSK_DETFO0` (no detector imaged the pixel), `MSK_QUALIT` (lost, degraded, defective, saturated) and `MSK_CLASSI` (opaque cloud, cirrus, snow/ice). Invalid pixels are set to 0 in `data` and flagged in the masks.

The acquisition as a whole is then accepted or rejected on measured fractions. Defaults: at least 99% of pixels valid in all bands, at most 20% no-data in the 32-pixel border ring, at most 1% saturated and 5% artefact samples; the in-square cloud filter is available but off by default, since the same information is recorded for filtering afterwards. The `status` column is `saved` or `rejected`, with `rejection_reason` giving the failing test.

## 5 metadata.csv

One row per granule returned by the search, 45 columns, in five groups:

- **Identification** — file name, date, MGRS tile, ESA scene cloud cover, product id, sensing time, platform, processing baseline, source CRS, AOI coverage.
- **Decision** — `status`, the stage that decided it, and the rejection reason.
- **Acquisition geometry** — relative orbit, orbit state, sun elevation and azimuth, view incidence angle, dataake id, processing time. These matter for time-series work: illumination and view angle change between dates.
- **Pixel statistics** — valid and no-data fractions overall and per band, edge no-data, per-band min/max/p02/p50/p98 (normalisation without re-reading the arrays).
- **Quality** — cloud, cirrus, snow/ice, artefact and saturation fractions, the masks actually read per band, and the quality-control configuration used, so a dataset built over several runs stays auditable.

## 6 Volume

A measured file of a cloud-free urban scene occupies 19.1 MB (13 bands plus all masks, compressed); complex or cloudy scenes compress differently, so 19 MB is a conservative planning figure. Per site, volume is  $19 \text{ MB} \times n_{\text{dates}}$ ; for the 3928-site WorldStrat list:

| Scenario                 | Dates per site | Total  |
|--------------------------|----------------|--------|
| One clear scene per site | 1              | 75 GB  |
| One per quarter, 1 year  | 4              | 300 GB |
| Monthly, 1 year          | 12             | 900 GB |
| Monthly, 2 years         | 24             | 1.8 TB |

Downloads use the CDSE S3 endpoint and read only the window covering the footprint, not the full 110 km granule. The catalogue rate-limits bursts of queries (HTTP 429), so batch runs pause between sites and retry with exponential back-off.

## 7 Known limitations

- **No atmospheric correction.** L1C is top-of-atmosphere; haze and aerosols are part of the signal. Use L2A, or correct downstream, if surface reflectance is required.
- **Clouds are flagged, not removed.** No gap filling or temporal interpolation is applied; masked pixels stay 0.
- **Cloud filtering biases the season.** `-max_cloud` selects the dry season, most strongly in the tropics. Widening the date range or filtering afterwards on the recorded per-scene statistics avoids over-selecting.
- **Irregular time steps.** Revisit is 5 days at best, but cloud and quality rejection leave uneven gaps; dates are not on a regular grid and differ between sites.
- **The ESA cloud percentage covers the whole granule** (up to  $100 \times 100 \text{ km}$ ), not the  $10.24 \text{ km}$  square. The in-square fractions in `metadata.csv` are the precise figure.
- **Footprint versus label.** With the WorldStrat list, the land-cover label describes the central 2.4% of the scene only; see the representativeness report for the sampling biases of that list.

## 8 Loading

```
z = np.load("data/site/2023-06-04_31UDQ.npz")
bands = z["bands"].tolist()
refl = (z["data"].astype("float32") - 1000) / 10000
refl[~z["valid_mask"]] = np.nan
```

# ['B01', 'B02', ...]  
# DN -> reflectance, baseline >= 04.00  
# drop unusable pixels